

DILLAN IMANS

Medical AI Researcher

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SUMMARY

Medical AI Researcher with first-author publications at EMA4MICCAI, IEEE BIBM, and Diagnostics, working on data- and label-efficient methods, including knowledge distillation, unsupervised domain adaptation, and unpaired cross-modal learning for diagnostic imaging and surgical vision.

EDUCATION

B.S. Computer Science & Engineering, Sungkyunkwan University **Aug 2022 - Jun 2026**
95.7 / 100 · Dean's List · Academic Excellence Scholarship

EXPERIENCE

AI Research Engineer · Superintelligence Lab SKKU, South Korea **Oct 2024 - Jun 2026**

- Built end-to-end retinal screening system (optic disc/cup segmentation & vessel analysis) and its backend; ONNX-optimized for low-latency inference, delivered to a clinical partner
- Developed a two-phase unsupervised domain adaptation framework (frequency alignment + SAM-guided refinement) for cross-modality brain tumor segmentation; +9.0 Dice over strongest baseline across 6 BraTS 2021 settings (First Author, IEEE BIBM 2025)
- Built a graph-mediated distillation framework for unpaired MRI-to-fundus hypertension prediction across disjoint cohorts; +5.2 AUC, +3.2 AUPRC over best baseline, MRI-free at inference (First Author, EMA4MICCAI 2026)
- Built a codebook-based distillation framework for unpaired cross-modal classification; +5.7 and +2.5 macro-F1 on OCT→fundus and CT→X-ray (First Author, under submission)

AI Researcher · Labren CUHK, Hong Kong **Jun 2024 - Jun 2026**

- Built preprocessing infrastructure and annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips)
- Curated a dataset and ran initial model experiments for a multimodal surgical robotics paper in collaboration with NVIDIA, contributing as a co-author
- Co-authored 3 manuscripts under submission in surgical vision and multimodal robotics

AI Researcher · Infolab SKKU, South Korea **Feb 2024 - Oct 2024**

- Explainable multi-layer dynamic ensemble for depression detection and severity assessment; 88.3% accuracy across 2,763 clinical records with SHAP interpretability, benchmarked against 20+ algorithms (First Author, Diagnostics 2024)

PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ

- Imans D. et al. "Clinical Graph-Mediated Distillation for Unpaired MRI-to-CFI Hypertension Prediction." EMA4MICCAI 2026.
- Imans D. et al. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." IEEE BIBM 2025.
- Imans D. et al. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." Diagnostics, 14(21), 2385 (2024).
- Imans D. et al. "Shared Semantic Codebook Distillation for Unpaired Cross-Modal Medical Classification." Preprint, under submission.
- 3 co-authored preprints under submission (surgical vision, multimodal robotics)

TECHNICAL SKILLS

Programming Languages: Python, C/C++

ML / AI: PyTorch, MONAI, HuggingFace, scikit-learn, OpenCV, wandb, TensorBoard

Medical: NiBabel, SimpleITK, pydicom

Tools: Git, Linux, Docker, Bash

Spoken Languages: English & Indonesian (Fluent), Mandarin & Korean (Conversational)